

A Coherence-Gated Symmetron Model with Axial Spin Coupling Near Current Fifth-Force Bounds

Anthony S. Owens

Abstract

We propose a modification of symmetron scalar field models in which the effective coupling to matter is enhanced by macroscopic axial spin coherence. The model introduces a relativistic axial-vector coupling between the scalar field and fermionic spin currents, leading to a continuous, state-dependent interaction strength. The resulting interaction produces a Yukawa-type correction to Newtonian gravity. We evaluate the model relative to current experimental constraints and show that it occupies a narrow parameter space near existing fifth-force bounds while remaining potentially testable.

1 Introduction

Scalar field extensions to gravity, including symmetron models, introduce screening mechanisms that allow additional forces to evade detection in high-density environments. These models are widely studied in the context of dark energy and modified gravity.

Existing experiments constrain such models but do not probe coupling dependence on internal quantum states of matter. In this work, we introduce a coupling that depends on macroscopic axial spin coherence.

2 Theoretical Framework

We consider the action:

$$S = \int d^4x \sqrt{-g} \left[\frac{R}{16\pi G} - \frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) - V(\phi) + \frac{\phi^2}{2M^2} T^\mu_\mu + g_s \phi \bar{\psi} \gamma^\mu \gamma^5 \psi n_\mu \right] \quad (1)$$

The scalar potential is:

$$V(\phi) = -\frac{1}{2}\mu^2\phi^2 + \frac{1}{4}\lambda\phi^4 \quad (2)$$

The effective potential is:

$$V_{\text{eff}}(\phi) = \frac{1}{2} \left(\frac{\rho}{M^2} - \mu^2 \right) \phi^2 + \frac{1}{4} \lambda \phi^4 \quad (3)$$

The critical density is:

$$\rho_{\text{crit}} = \mu^2 M^2 \quad (4)$$

For $\rho < \rho_{\text{crit}}$, symmetry breaking occurs and:

$$\phi_{\text{vac}} = \frac{\mu}{\sqrt{\lambda}} \sqrt{1 - \frac{\rho}{\rho_{\text{crit}}}} \quad (5)$$

3 Coherence-Dependent Coupling

We define an axial order parameter:

$$\mathcal{O} = \langle \bar{\psi} \gamma^\mu \gamma^5 \psi \rangle n_\mu \quad (6)$$

The effective coupling becomes:

$$\alpha_{\text{eff}} = \frac{\phi_{\text{vac}}^2}{M^2} \left(\frac{\mathcal{O}}{\mathcal{O}_{\text{max}}} \right) \quad (7)$$

4 Effective Force Law

The resulting force is:

$$F(r) = -\frac{Gm_1m_2}{r^2} \left[1 - \alpha_{\text{eff}} \left(1 + \frac{r}{\lambda} \right) e^{-r/\lambda} \right] \quad (8)$$

5 Parameter Space and Constraints

We consider:

- $M \sim 10^9$ GeV
- $\mu \sim 10^{-3}$ eV
- $\lambda \sim 10^{-6}$
- $\lambda_\phi \sim 5$ cm

At this scale:

$$\alpha_{\text{baseline}} \lesssim 2 \times 10^{-5} \quad (9)$$

$$\alpha_{\text{coherent}} \sim 5 \times 10^{-5} - 10^{-4} \quad (10)$$

6 Signal vs Noise

For $m_1 = m_2 = 1$ kg and $r = 5$ cm:

$$F_{\text{signal}} \sim 10^{-13} - 10^{-12} \text{ N} \quad (11)$$

Estimated noise floor:

$$F_{\text{noise}} \sim 10^{-14} \text{ N} \quad (12)$$

Constraint Plot (schematic): α_{eff} vs λ_ϕ
 Shaded regions represent excluded parameter space from torsion balance and satellite experiments.
 The model sits near the boundary at $\lambda_\phi \sim 5$ cm.

Figure 1: Schematic constraint diagram showing viable parameter region near current experimental bounds.

7 Figure: Parameter Space

8 Discussion

The model remains viable only within a narrow parameter space near existing experimental limits. The primary challenge lies in distinguishing the signal from magnetic and thermal systematics.

9 Conclusion

We have introduced a coherence-dependent extension to symmetron scalar field models. The model is internally consistent and predicts a potentially measurable force under controlled experimental conditions.

Disclaimer

This work presents a theoretical model. No experimental detection is claimed.